

HK IMO Training 2020-2021
Problem Set 19
For 12 June 2021



Problem 73. Show that for each positive integer n ,

$$n! = \prod_{i=1}^n \text{lcm}(1, 2, \dots, \lfloor n/i \rfloor).$$

Problem 74. Prove that for any positive integers $a_1, a_2, \dots, a_n$,

$$\prod_{1 \leq i < j \leq n} \frac{a_i - a_j}{i - j}$$

is an integer.

Problem 75. Prove that for any positive integer k ,

$$(k^2)! \times \prod_{j=0}^{k-1} \frac{j!}{(j+k)!}$$

is an integer.

Problem 76. Find all positive integers (x, y, n) such that

$$x^n - y^n = x! - y!.$$